

Yann Baglin-Bunod

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EDUCATION

University of California, San Diego

M.S. in Electrical & Computer Engineering – Machine Learning & Data Science

B.S. in Electrical Engineering — GPA: 3.75/4.0

San Diego, CA

Jan. 2026 – Dec. 2026

Sep. 2021 – Jun. 2025

École Polytechnique, IP Paris

Global Exchange Program – Ranked #1 Engineering School in France

Paris, France

Aug. 2023 – Mar. 2024

WORK EXPERIENCE

OpenAirlines

Data Science Intern

Jun. 2025 – Aug. 2025

Toulouse, France

- Designed an end-to-end natural language processing pipeline converting analyst queries into structured JSON analyses over flight operations data
- Shipped a prototype reaching >95% accuracy on a 200+ question validation set, enabling non-technical users to query flight data directly
- Tracked experiments and ablations with MLflow to compare prompt strategies and pipeline configurations against a held-out benchmark

UC San Diego – Summer Research Internship Program

Medical Swallow Analysis Project

May 2024 – Sep. 2024

San Diego, CA

- Applied deep learning to medical diagnostics; engineered data augmentation and an audio segmenter to preprocess swallowing sounds for classification
- Built a CNN segmenter and SVM classifier over 8,459 labeled sEMG segments, running MLflow-tracked ablations with leakage-checked splits

UC San Diego – Department of Electrical & Computer Engineering

Instructional Assistant, ECE 15: C Programming

Jan. 2023 – Dec. 2023

San Diego, CA

- Tutored 140 students in C programming across weekly office hours; earned top teaching feedback over two consecutive quarters

PROJECTS & RESEARCH

Gaussian Process Fusion Research (*manuscript in preparation*)

Python, PyTorch, Gemini embeddings

- Ran a 9,703-config ablation of multimodal Gaussian Process fusion (9 embedders $\times$ 3 scaling methods $\times$ 5 ensemble modes); a data-pipeline fix raised best test R^2 from 0.629 to 0.662
- Built a spectrally-normalized MLP over Gemini text embeddings to extract a 128-d GP input, reaching test R^2 0.6685 with calibrated uncertainty, beating all 9,703 raw-feature configs

Hybrid RAG over Technical Documentation

Python, FAISS, BM25, bge-m3, HyDE, cross-encoder

- Engineered a hybrid retrieval-augmented generation pipeline (hierarchical chunking, bge-m3 dense + BM25 sparse fused by reciprocal-rank fusion, cross-encoder rerank), reaching F1@5 0.853 / Precision@5 0.889
- Designed a HyDE query stage driving retrieval from a hypothetical-answer embedding, with parent-chunk expansion and a 1,700-token context packer

Fine-Tuning Small Language Models for Summarization

Python, Hugging Face, LoRA, LLM-as-judge

- Fine-tuned three small LLMs (GPT-Neo-125M full; Phi-1.5B, TinyLlama-1.1B via LoRA) on 5,000 Wikipedia bios; GPT-Neo-125M led every metric (ROUGE-1 0.67, BLEU 0.32), beating both larger models
- Implemented an LLM-as-judge loop where Mistral-7B-Instruct scored each summary on relevance, coherence, and conciseness, fully automating evaluation

On-Device Snoring Detection App

Kotlin, TensorFlow Lite, Qualcomm AI Hub, CNN

- Developed a real-time snore detection app with a lightweight CNN-based TensorFlow Lite model deployed on Android for on-device inference; profiled and optimized runtime via Qualcomm AI Hub

SKILLS

Languages: Python, C, C++17, Kotlin, MATLAB, Assembly

Libraries & Frameworks: PyTorch, TensorFlow, TensorFlow Lite, Scikit-learn, NumPy, Pandas, Hugging Face, FAISS

ML & DS: Deep Learning (CNNs, RNNs, Transformers), Statistical Learning, NLP, RAG, LLMs, Computer Vision, Reinforcement Learning, On-device Inference

Tools & Platforms: MLflow, CUDA, Qualcomm AI Hub, RapidFire AI, Git, Docker, Jupyter, Linux

Coursework: Statistical Learning (SVMs, GPs, kernel methods), Optimization & Acceleration of Deep Learning, Linear Algebra & Applications, Software Foundations I & II (C++17), Vision Transformers, Self-Supervised Learning, Generative AI (Stable Diffusion, LoRA, RLHF), Reinforcement Learning (DQN, MDPs, Bandits)